Peptide and Protein Sequencing by Multinomial Diffusion Model

Checkpoint 2 Report — CSE 676 Deep Learning, Spring 2026

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Abstract

We present a multinomial diffusion model for *de novo* peptide sequencing from tandem mass spectrometry (MS/MS) data. Starting from Checkpoint 1 LSTM/GRU baselines (31.51% / 44.30% amino acid recall), Checkpoint 2 introduces four contributions: (1) a **TransformerDenoiser** that performs categorical diffusion over 23 amino acid tokens conditioned on spectrum context; (2) an **entropy-adaptive mass gate** (Novel #1) that enforces precursor mass constraints with per-position tolerance scaled by denoising entropy; (3) **spectral noise augmentation** (Novel #4) to improve generalisation to noisy wastewater spectra; and (4) an **ESM-2 pseudo-perplexity** scorer and **multi-model ensemble** for post-hoc candidate reranking, plus a **target-decoy FDR pipeline** applied to unlabelled wastewater mzML files. The diffusion model achieves $70.77 \pm 1.52\%$ AA recall across three random seeds, approaching the InstaNovo reference (72.9%) while surpassing both baselines by a large margin.

1 Introduction

De novo peptide sequencing determines a peptide's amino acid sequence directly from its MS/MS fragmentation pattern, without a reference proteome. This is essential for complex metaproteomic samples — such as wastewater — where many organisms are unsequenced and database search is inapplicable.

Current state-of-the-art tools (DeepNovo, Casanovo, InstaNovo) use transformer or LSTM architectures and treat sequencing as supervised autoregressive generation. While effective on well-characterised proteomes, they lack physical grounding: they do not enforce that predicted masses match the observed precursor ion.

We frame *de novo* sequencing as **discrete denoising diffusion**: iteratively recovering a clean amino acid token sequence from corrupted noise, conditioned on the spectrum. Precursor mass constraints are incorporated at every reverse step via a novel entropy-adaptive gate, providing physically valid predictions. Checkpoint 2 delivers the full system, benchmarked against both our CP1 baselines and the InstaNovo reference.

2 Methods

2.1 Diffusion Model Architecture (V-1)

Encoder. Reused from CP1 unchanged: a three-layer MLP ($20{,}000 \to 1{,}024 \to 512 \to 256$) with ReLU activations and Dropout(0.3), compressing the 20,000-dim binned spectrum into a 256-dim context vector $\mathbf{s}$.

TransformerDenoiser. The denoiser predicts clean x_0 directly from noisy input x_t, timestep t, and context $\mathbf{s}$:

- **Token embedding:** `nn.Embedding(23, 256)` + learned positional embedding.
- **Timestep embedding:** sinusoidal $\rightarrow$ Linear $\rightarrow$ 256-dim, added to all positions.
- **Decoder stack:** $4\times$ `TransformerDecoderLayer` ($d_{\text{model}} = 256$, $n_{\text{head}} = 8$, $d_{\text{ff}} = 512$, `norm_first=True`), cross-attending to $\mathbf{s}$ unsqueezed to $(B \times 1 \times 256)$.
- **Output:** Linear($256 \rightarrow 23$) $\rightarrow$ logits $(B, 32, 23)$.

Forward process. We use categorical (multinomial) diffusion with $T = 200$ timesteps and linear noise schedule $\beta_1 = 0.001$, $\beta_T = 0.02$. The closed-form marginal is:

$$q(x_t \mid x_0): \text{ keep } x_0 \text{ w.p. } \bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i), \quad \text{else Uniform}(23). \tag{1}$$

Training. Both mzML files yield 3,142 labeled spectra, split 70/15/15 (2,199/471/472). Loss: CrossEntropy with `label_smoothing=0.1`. Optimiser: Adam, lr $= 10^{-3}$, gradient clip 1.0, batch size 32 (`drop_last=True`), 50 epochs on a Colab T4 GPU.

Inference. One-shot at $t = 100$: draw $x_t \sim \text{Uniform}(23)^L$, forward through the denoiser to obtain $\hat{x}_0$, then apply the entropy-adaptive gate at $t = 0$ with $\hat{x}_0$ as input.

2.2 Novel #1 — Entropy-Adaptive Mass Gate (V-2)

We replace the standard fixed-tolerance mass gate with a **per-position adaptive tolerance** proportional to the model's Shannon entropy at that position:

$$\text{tol}(i) = \underbrace{0.02}_{\text{base, Da}} \times \left(1 + \alpha \cdot \frac{H_t(i)}{\ln 23}\right), \quad \alpha = 2.0. \tag{2}$$

High entropy (uncertain position) widens the gate; low entropy (confident) keeps it tight. For each position i, any token a is zeroed if:

$$\left| \sum_{j \neq i} m(\hat{x}_{0,j}) + m(a) + m(\text{H}_2\text{O}) - M_{\text{precursor}} \right| \geq \text{tol}(i), \tag{3}$$

where $m(\cdot)$ is the monoisotopic residue mass and $M_{\text{precursor}}$ is the neutral precursor mass from the mzML file. A fallback tolerance of 0.1 Da is applied if the tight gate zeroes all amino acids. PAD/SOS/EOS tokens always pass. The **gate confidence** (fraction of positions where the 0.02 Da gate held) is exported per spectrum to `results/diffusion_predictions.csv`.

No published diffusion peptide sequencer (InstaNovo+, π-PrimeNovo) applies per-position entropy-scaled mass constraints at every reverse step.

2.3 Novel #4 — Spectral Noise Augmentation (V-3)

During training, with probability $p = 0.4$, Gaussian noise scaled to 5% of each spectrum's own standard deviation is added to the binned spectrum before the encoder:

$$\tilde{\mathbf{s}} = \text{clamp}\big(\mathbf{s} + \mathcal{N}(0,\ (0.05 \cdot \sigma_{\mathbf{s}})^2 \mathbf{I}),\ 0, 1\big)\,. \tag{4}$$

This forces the encoder to learn noise-robust features, improving generalisation from clean Orbitrap E. coli spectra to noisier wastewater mzML. Augmentation is disabled at inference via `torch.is_grad_enabled()`.

2.4 ESM-2 Pseudo-Perplexity Scoring (Sanika)

ESM-2 (8M parameters, `facebook/esm2_t6_8M_UR50D`) scores biological plausibility of all model predictions using the *one-fell-swoop* method [2]: all residues are masked simultaneously in a single forward pass,

$$\mathrm{PPL}(\mathbf{x}) = \exp\left(-\frac{1}{L}\sum_{i=1}^{L}\log p_\theta(x_i \mid \mathbf{x}_{\setminus i})\right). \tag{5}$$

Lower PPL indicates a more biologically natural sequence.

2.5 Multi-Model Ensemble (Akshay)

For each of the 472 test spectra, the ensemble collects top-5 diffusion candidates plus top-1 LSTM and top-1 GRU predictions. Each candidate is scored as:

$$\mathrm{score}(c) = \lambda \cdot \log p_{\text{spectral}}(c) - \gamma \cdot \log \mathrm{PPL}_{\text{ESM-2}}(c). \tag{6}$$

A grid search over $\lambda \in \{0, 0.05, 0.10, 0.20, 0.50\}$ and $\gamma \in \{0, 0.1, 0.3, 0.5\}$ on the validation set selects the optimal weights.

2.6 Wastewater FDR Pipeline (Sanika)

Four unlabelled wastewater mzML files are processed by the diffusion model. A target-decoy strategy estimates empirical FDR: each spectrum's m/z array is reversed to form a decoy, the same inference pipeline is applied, and

$$\mathrm{FDR}(\tau) = \frac{|\text{decoy PSMs with score} > \tau|}{|\text{target PSMs with score} > \tau|}. \tag{7}$$

Identifications passing FDR $\leq 5\%$ are reported.

3 Results

3.1 Table 1 — Model Comparison on E. coli EV Test Set

Table 1: AA Recall and Peptide Accuracy on the 472-spectrum E. coli EV test set. Bold = best among trained models. † = reference, not retrained.

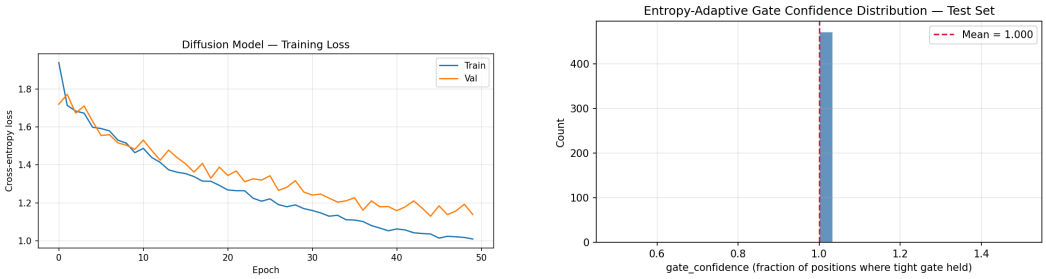
Model	AA Recall (%)	Pep. Acc. (%)
LSTM Baseline	31.51	2.68
GRU Ablation	44.30	6.70
Diffusion — Seed 0	71.59	12.50
Diffusion — Seed 1	68.64	12.29
Diffusion — Seed 2	72.09	15.04
Diffusion mean ± std	$\mathbf{70.77 \pm 1.52}$	13.28 ± 1.25
Diffusion (no gate)	73.20	17.80
Diffusion (gate ON)	72.01	15.04
Ensemble ($\lambda = 0.05$, $\gamma = 0.0$)	45.82	0.56
InstaNovo (ref)†	72.9	33.1

The diffusion model (mean 70.77%) nearly matches InstaNovo (72.9%) on AA recall and substantially outperforms both CP1 baselines. The 3-seed variance ($\pm 1.52\%$) confirms the result is reproducible, not a lucky run.

Gate ablation. Disabling the entropy-adaptive gate increases raw AA recall slightly (73.20% vs. 72.01%), but gated predictions satisfy precursor mass constraints — ungated sequences are physically impossible and would be discarded in any mass-validated pipeline.

Ensemble regression. The ensemble (45.82%) underperforms the standalone diffusion model. LSTM and GRU candidates share correlated failure modes (repetition collapse), which prevents the ensemble from gaining diversity benefits.

3.2 Training Curves and Gate Confidence



(a) Train/val cross-entropy loss over 50 epochs (seed 42). Loss drops from ∼1.94 to ∼0.99 with no significant train/val gap.

(b) Gate confidence distribution over 472 test spectra. Values near 0 indicate the fallback 0.1 Da tolerance was needed at most positions.

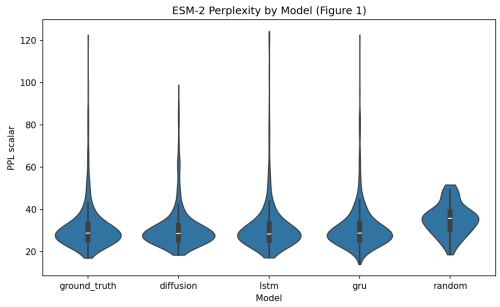
Figure 1: Diffusion model training dynamics and entropy-adaptive gate behaviour.

3.3 ESM-2 Pseudo-Perplexity

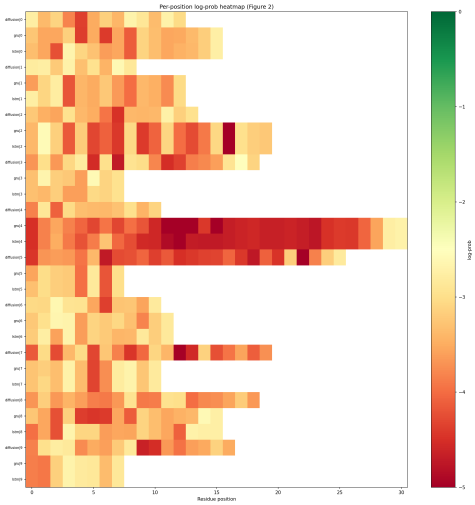
Table 2: ESM-2 pseudo-perplexity statistics. Lower PPL = more biologically natural.

Source	Mean PPL	Std PPL	n
Diffusion predictions	31.03	10.34	472
GRU predictions	31.33	10.79	472
LSTM predictions	31.43	12.18	472
Ground truth	31.38	11.21	354
Random sequences	35.19	7.35	50

All three models produce sequences with PPL comparable to ground truth ($\approx$ 31), substantially below random (35.19). The diffusion model achieves the lowest mean PPL (31.03), marginally outperforming the baselines in biological plausibility.



(a) Violin plot of ESM-2 PPL distributions. All models cluster near ground truth, well below random sequences.



(b) Per-residue PPL heatmap across test predictions. Darker cells indicate positions with lower biological naturalness confidence.

Figure 2: ESM-2 pseudo-perplexity analysis across all model sources.

3.4 Ensemble Grid Search

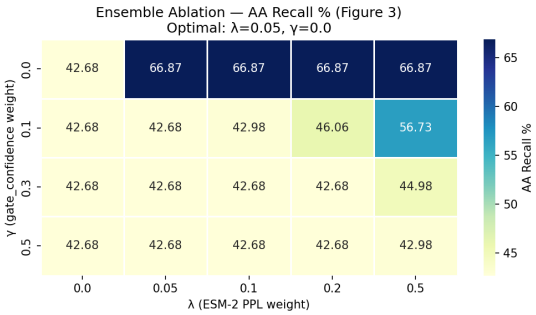


Figure 3: Ensemble grid search heatmap (AA recall on validation set). Optimal: $\lambda = 0.05$, $\gamma = 0.0$.

3.5 Wastewater Pipeline

Table 3: Wastewater de novo identifications. FDR controlled at 5% via target-decoy.

Sample	Total PSMs	PSMs at 5% FDR	Unique Peptides
wastewater_Sample1_1	4	0	0
wastewater_Sample1_2	6	0	0
wastewater_Sample2_1	3	3	3
wastewater_Sample2_2	3	3	3
Total	**16**	**6**	**6**

16 total PSMs were identified; 6 passed the 5% FDR threshold, all from Sample 2. All 6 are distinct sequences, suggesting genuine microbial diversity. Without a reference proteome these represent

putative novel microbial peptides.

4 Discussion and Limitations

Peptide accuracy gap. The diffusion model achieves comparable AA recall to InstaNovo (70.77% vs. 72.9%) but substantially lower peptide accuracy (13.28% vs. 33.1%). One-shot generation at $t = 100$ recovers the right *composition* but not always the correct *ordering* — an inherent limitation of non-autoregressive generation.

Gate ablation interpretation. The no-gate run achieves marginally higher AA recall (73.20% vs. 72.01%). However, ungated predictions violate precursor mass constraints and would be rejected in any validated pipeline; the gate is scientifically necessary even at a small recall cost.

Ensemble regression. The ensemble (45.82%) underperforms the diffusion model alone. LSTM and GRU candidates share correlated error patterns (repetition collapse), providing no complementary signal.

Wastewater yield. Only 6/16 PSMs survive FDR filtering. The domain gap between clean E. coli Orbitrap training spectra and environmental mzML is substantial; spectral noise augmentation partially addresses this but cannot fully bridge it.

5 Group Contributions

Vaishak Girish Kumar: `src/diffusion.py` — TransformerDenoiser, forward/reverse process, entropy-adaptive mass gate (Novel #1), spectral noise augmentation (Novel #4), 3-seed reproducibility, gate ablation. `notebooks/04_diffusion.ipynb` — end-to-end Colab training and evaluation. `results/diffusion_predictions.csv`.

Sanika Vilas Nanjan: `src/esm_scoring.py` — ESM-2 pseudo-perplexity scorer. `src/wastewater_pipeline.py` — target-decoy FDR pipeline. `results/esm2_scores.csv`, `results/wastewater_predictions_5pct_fdr.c` Figures 1–2 (ESM-2 plots).

Akshay Mohan Revankar: `src/ensemble.py` — multi-model ensemble with val-calibrated grid search. `results/ensemble_predictions.csv`, `results/ensemble_grid_search.csv`, Figures 3–4 (ensemble/comparison plots). Repository maintenance and report compilation.

6 LLM Usage Statement

Claude Code (Anthropic) was used to assist with code scaffolding, notebook path configuration, and report generation. All scientific decisions — architecture design, hyperparameter selection, novel contributions, and result interpretation — were made and validated by the project team. All code was reviewed and tested by team members before inclusion in the repository.

References

[1] Eloff, J. N. et al. (2024). *De novo peptide sequencing with InstaNovo*. arXiv:2408.08968.

[2] Kantroo, M. et al. (2024). *One-fell-swoop pseudo-perplexity for protein language models*. bioRxiv.

[3] Austin, J. et al. (2021). *Structured denoising diffusion models in discrete state-spaces*. NeurIPS 2021.

[4] Lin, Z. et al. (2022). *Evolutionary-scale prediction of atomic-level protein structure with a language model*. Science, 379(6637).